



**SIMATS**  
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

## **CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM**

### **EXPERIMENT NO: 04**

**Name: Avishek N**

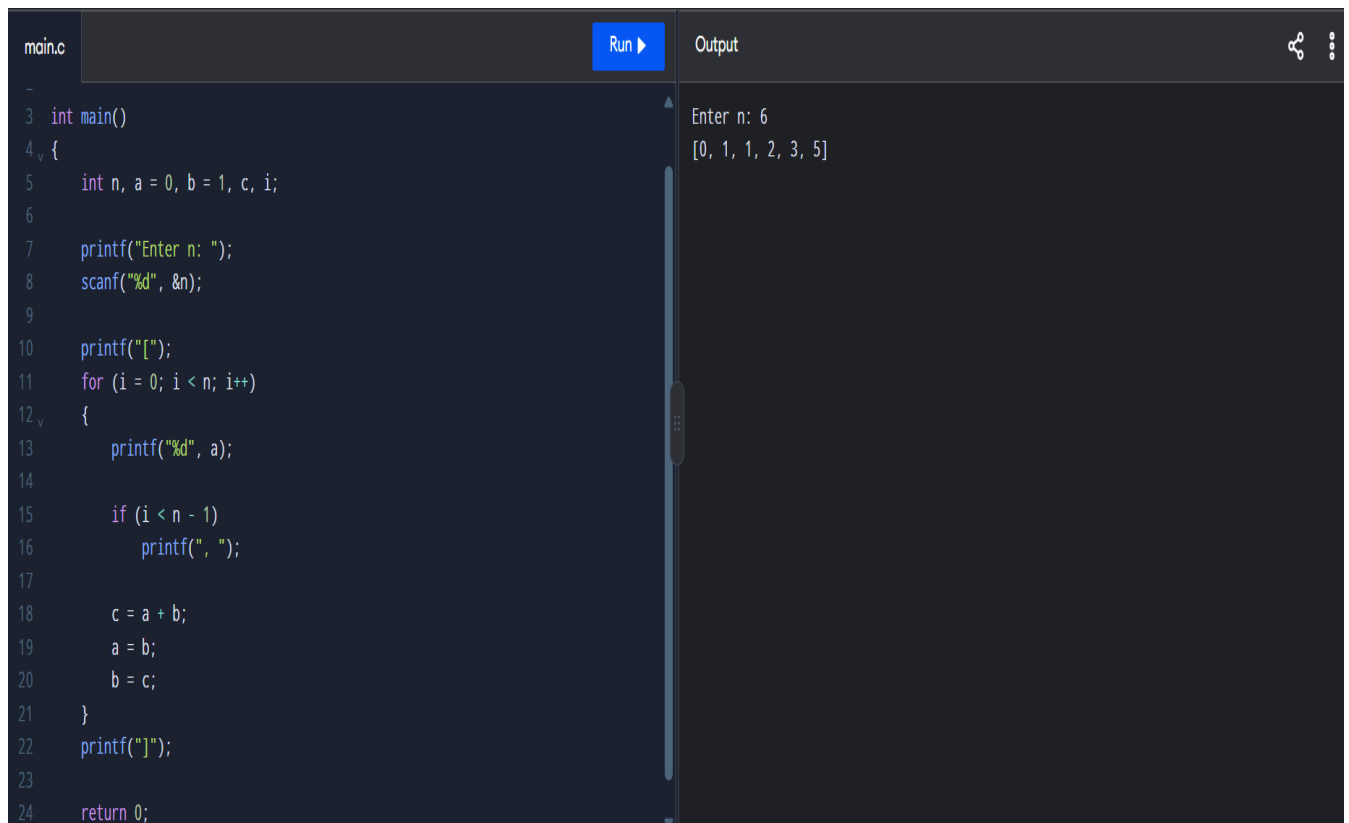
**Reg No: 192511009**

#### **Program:**

```
#include <stdio.h>

int main()
{
    int n, a = 0, b = 1, c, i;
    printf("Enter n: ");
    scanf("%d", &n);
    printf("[");
    for (i = 0; i < n; i++)
    {
        printf("%d", a);
        if (i < n - 1)
            printf(", ");
        c = a + b;
        a = b;
        b = c;
    }
    printf("]");
    return 0;
}
```

## Output:



The image shows a code editor window with a file named 'main.c'. The code is a C program that calculates the first 'n' terms of a Fibonacci sequence. The program prompts the user to 'Enter n:' and then prints the sequence. The output window shows the user input '6' and the resulting sequence '[0, 1, 1, 2, 3, 5]'. The code is as follows:

```
main.c Run ▶ Output
-
3 int main()
4 {
5     int n, a = 0, b = 1, c, i;
6
7     printf("Enter n: ");
8     scanf("%d", &n);
9
10    printf("[");
11    for (i = 0; i < n; i++)
12    {
13        printf("%d", a);
14
15        if (i < n - 1)
16            printf(", ");
17
18        c = a + b;
19        a = b;
20        b = c;
21    }
22    printf("]");
23
24    return 0;
```

Enter n: 6  
[0, 1, 1, 2, 3, 5]